

STAVANGER VAALAND, Norway

WMO: 014160

Lat: 58.9574N

Lon: 5.7287E

Elev: 236

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.9	23.6	6.7	7.8	26.5	10.4	9.4	26.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	76.2	62.2	72.1	60.5	68.7	59.0	64.0	72.8	62.3	69.2	60.6	66.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	80.5	66.1	59.3	75.9	64.8	57.8	71.9	63.4	29.4	73.0	28.1	69.4	26.9	66.4	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	20.1	83.1	4.5	5.1	16.9	86.8	14.3	89.8	11.7	92.7	8.5	96.4	(4)
(5)			WB	18.1	66.8	4.8	2.2	14.7	68.4	11.9	69.6	9.2	70.9	5.7	72.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	35.5	35.6	39.3	45.6	51.2	55.6	60.3	59.8	55.9	48.4	42.4	38.1	(6)	
	DBStd	10.26	5.53	5.61	4.66	5.32	6.66	4.54	5.19	3.88	4.32	4.52	5.58	6.18	(7)	
	HDD50	2090	450	403	332	155	62	3	0	0	5	80	232	368	(8)	
	HDD65	6477	915	823	797	582	434	286	168	169	277	514	679	833	(9)	
	CDD50	1132	0	0	0	22	98	172	319	305	181	32	3	0	(10)	
	CDD65	43	0	0	0	4	5	23	9	2	0	0	0	0	(11)	
	CDH74	221	0	0	0	0	26	25	134	28	8	0	0	0	(12)	
	CDH80	34	0	0	0	0	3	1	28	2	0	0	0	0	(13)	
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)	
(15) Precipitation	PrecAvg	47.2	4.1	3.0	2.8	2.2	2.6	2.7	3.5	4.8	5.6	5.7	5.5	4.7	(15)	
	PrecMax	68.2	9.3	7.0	7.9	5.0	6.6	6.3	9.6	9.5	12.9	11.8	9.6	9.8	(16)	
	PrecMin	33.1	0.2	0.0	0.1	0.1	0.1	0.6	0.8	0.2	1.1	1.5	1.0	0.3	(17)	
	PrecStd	7.6	2.4	1.7	1.8	1.0	1.4	1.3	2.0	2.0	2.2	2.4	2.2	2.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	48.5	53.6	67.0	77.3	77.9	83.4	78.1	72.9	64.1	56.7	50.4	(19)	
		MCWB	44.3	43.7	46.6	51.9	61.0	61.0	65.6	62.5	61.7	56.9	51.3	48.2	(20)	
	2%	DB	45.6	45.8	49.3	61.7	71.9	70.7	77.4	72.0	67.4	59.4	53.1	48.9	(21)	
		MCWB	43.8	42.9	43.9	49.4	57.4	59.5	63.2	60.7	57.7	53.9	49.9	46.9	(22)	
	5%	DB	44.5	44.4	46.7	57.7	66.6	66.3	73.1	68.9	64.8	56.5	50.9	47.4	(23)	
		MCWB	42.7	42.1	42.6	47.8	55.1	56.8	62.0	59.5	56.8	51.7	47.9	45.3	(24)	
	10%	DB	43.2	43.1	45.1	54.1	62.0	63.1	68.8	66.4	62.3	54.5	49.7	46.0	(25)	
		MCWB	41.2	40.9	41.5	46.1	52.7	55.4	60.5	58.5	56.0	50.3	46.9	43.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.9	44.9	48.2	52.7	62.2	62.9	67.2	64.8	63.4	58.0	53.0	49.2	(27)	
		MCDB	46.4	46.4	51.5	62.7	75.1	71.9	79.3	72.2	71.6	61.3	54.1	49.9	(28)	
	2%	WB	44.5	43.7	45.3	50.4	59.0	60.3	65.0	62.6	60.2	55.4	50.8	47.4	(29)	
		MCDB	45.2	45.1	48.1	59.1	69.6	69.2	75.3	68.9	64.3	58.5	52.5	48.6	(30)	
	5%	WB	43.1	42.6	43.7	49.0	56.1	58.2	63.3	61.1	58.3	53.2	49.2	45.8	(31)	
		MCDB	44.0	44.1	46.1	56.1	64.8	65.1	71.2	66.1	63.3	55.6	50.8	47.1	(32)	
	10%	WB	41.3	41.1	42.4	47.2	53.7	56.4	61.5	59.7	56.6	51.4	47.2	43.9	(33)	
		MCDB	42.7	42.8	44.4	52.9	61.0	62.0	67.9	64.8	61.0	53.5	49.1	45.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.3	8.9	12.1	12.8	12.4	11.9	11.0	10.0	8.4	6.7	6.2	(35)	
		MCDBR	7.1	7.5	10.2	17.1	19.7	17.5	18.4	14.4	13.3	10.6	7.2	6.4	(36)	
	5% WB	MCWBR	7.1	6.2	6.7	8.7	9.5	8.3	7.7	6.8	6.8	7.0	6.4	6.9	(37)	
		MCWBR	7.0	7.2	9.4	15.3	18.9	15.7	16.6	12.2	11.9	8.6	7.3	7.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.324	0.344	0.376	0.366	0.360	0.374	0.374	0.355	0.335	0.305	0.277	(40)	
		taud	2.177	2.282	2.321	2.299	2.418	2.464	2.443	2.441	2.483	2.428	2.263	2.120	(41)	
	Ebn at Noon		157	219	251	262	275	278	271	261	248	215	157	121	(42)	
		Edh at Noon	15	22	29	35	33	32	32	30	25	19	14	11	(43)	
(44) All-Sky Solar Radiation	RadAvg		95	285	675	1213	1625	1753	1555	1200	770	370	131	61	(44)	
	RadStd		16	49	84	127	187	196	159	103	69	47	19	8	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A

Nomenclature: See separate page